## Experiment 9: write python program to Travelling Salesman Problem

**Aim:**

To write a Python program to implement the Travelling Salesman Problem (TSP) and find the minimum travel cost.

**Algorithm:**

1. Create a distance matrix for the cities.
2. Start from city A.
3. Visit every other city exactly once.
4. Return to the starting city.
5. Find the minimum total distance.

**Program:**

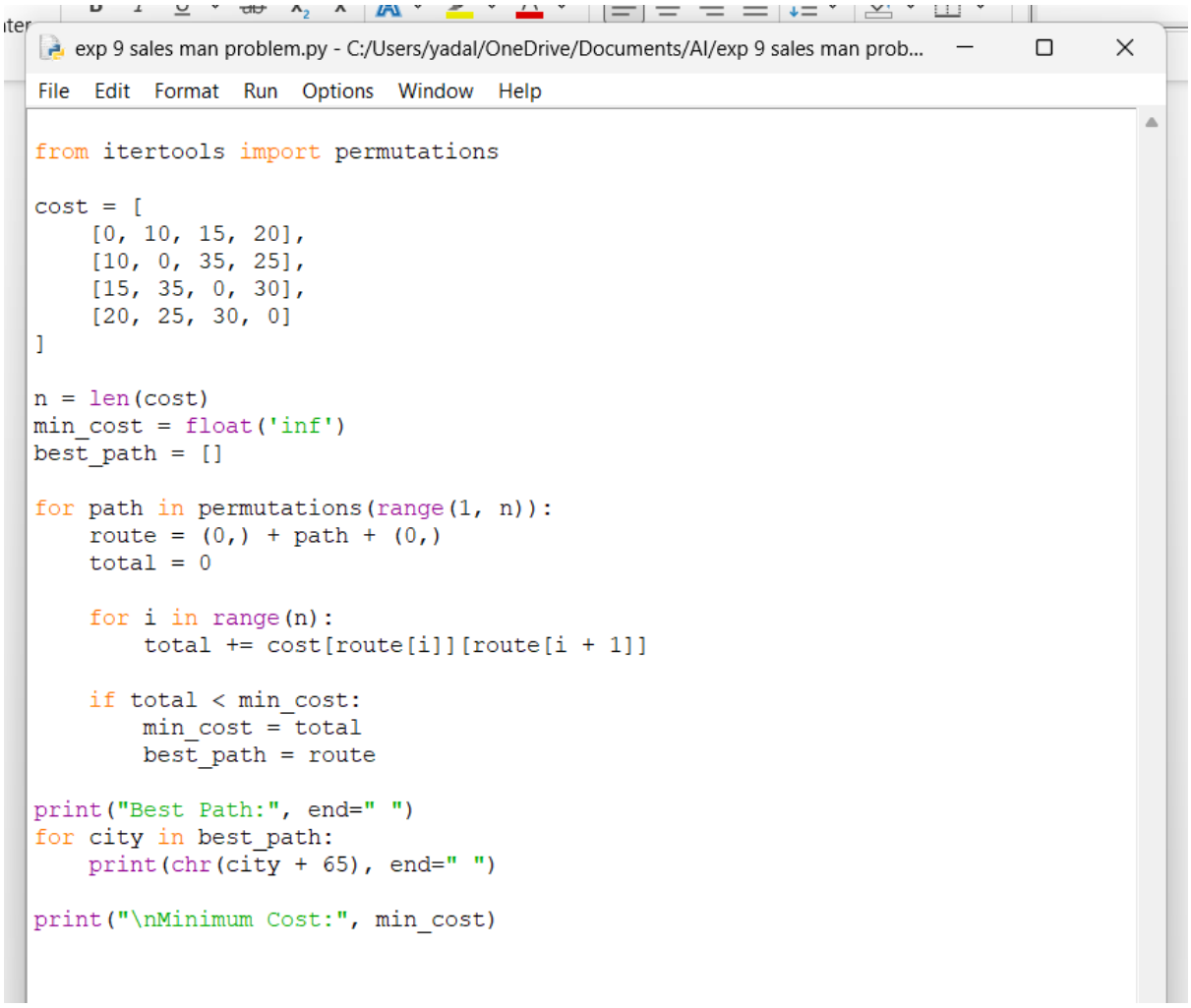
```
from itertools import permutations
cost = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]
n = len(cost)
min_cost = float('inf')
best_path = []
for path in permutations(range(1, n)):
    route = (0,) + path + (0,)
    total = 0
    for i in range(n):
        total += cost[route[i]][route[i + 1]]
    if total < min_cost:
        min_cost = total
        best_path = route
```

print("Best Path:", end=" ")

for city in best_path:

    print(chr(city + 65), end=" ")

print("\nMinimum Cost:", min_cost)




Output:

```
=== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 9 sales man problem.py ===
Best Path: A B D C A
Minimum Cost: 80
```

**Result:**

Thus, the Python program to implement the Travelling Salesman Problem was executed successfully, and the minimum travel cost was found to be 80.